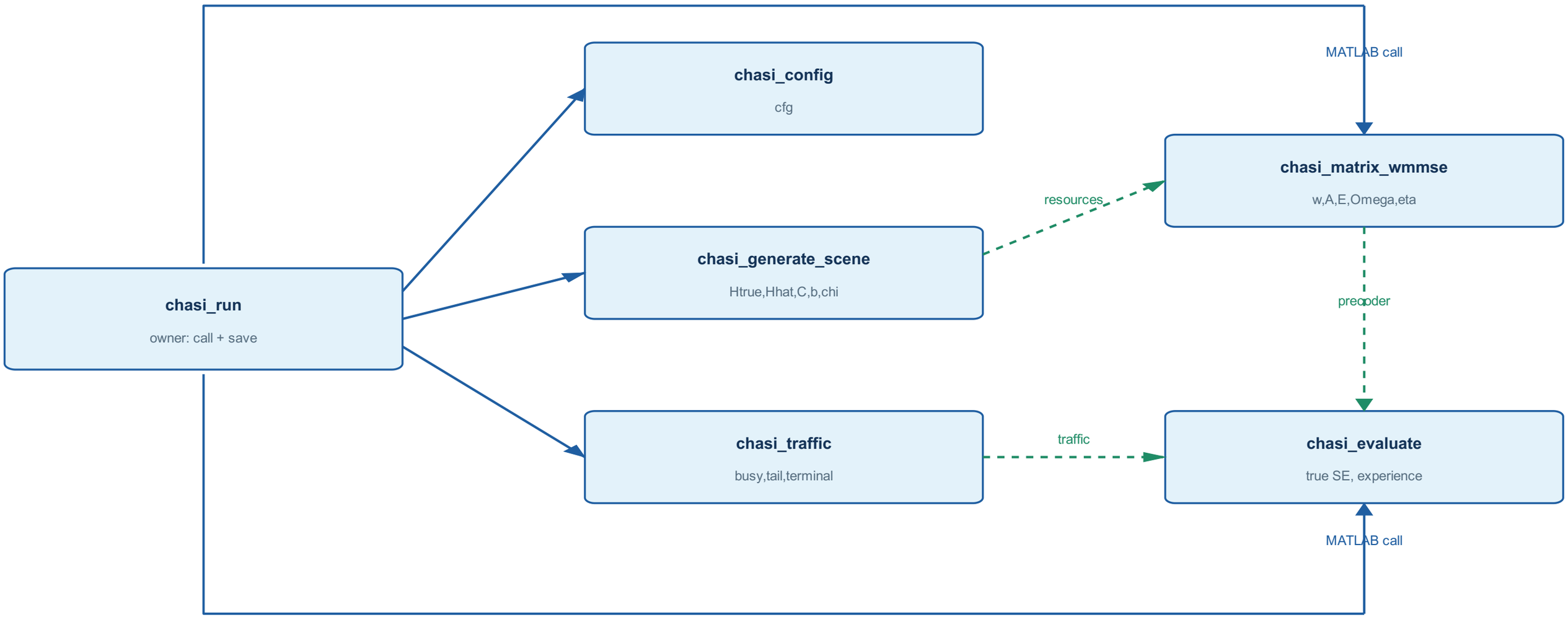
# 01 Actual MATLAB calls

MATLAB source: matlab/chasi/ at 9527089; formula source: docs/CHASI_UNIFIED_TRACE.tex

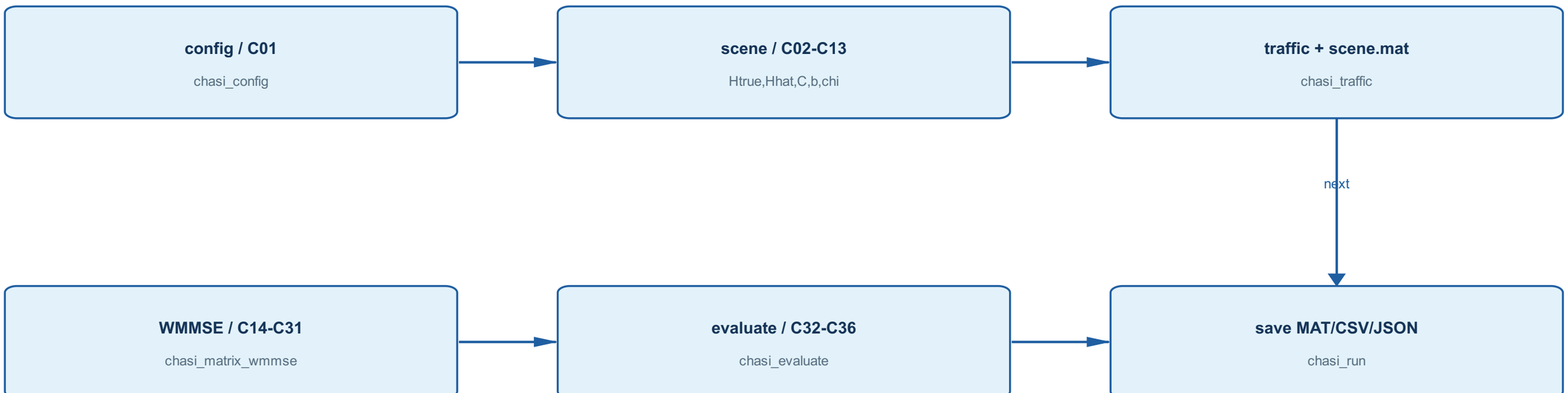
Blue solid = chasi_run call; green dashed = arrays passed to solver/evaluator.




Call evidence: chasi_run.m:39-45,90-97. chasi_run saves scene.mat before optimization, then precoder MAT, per-UE CSV and summary JSON.

# 02 Formal native MATLAB chain

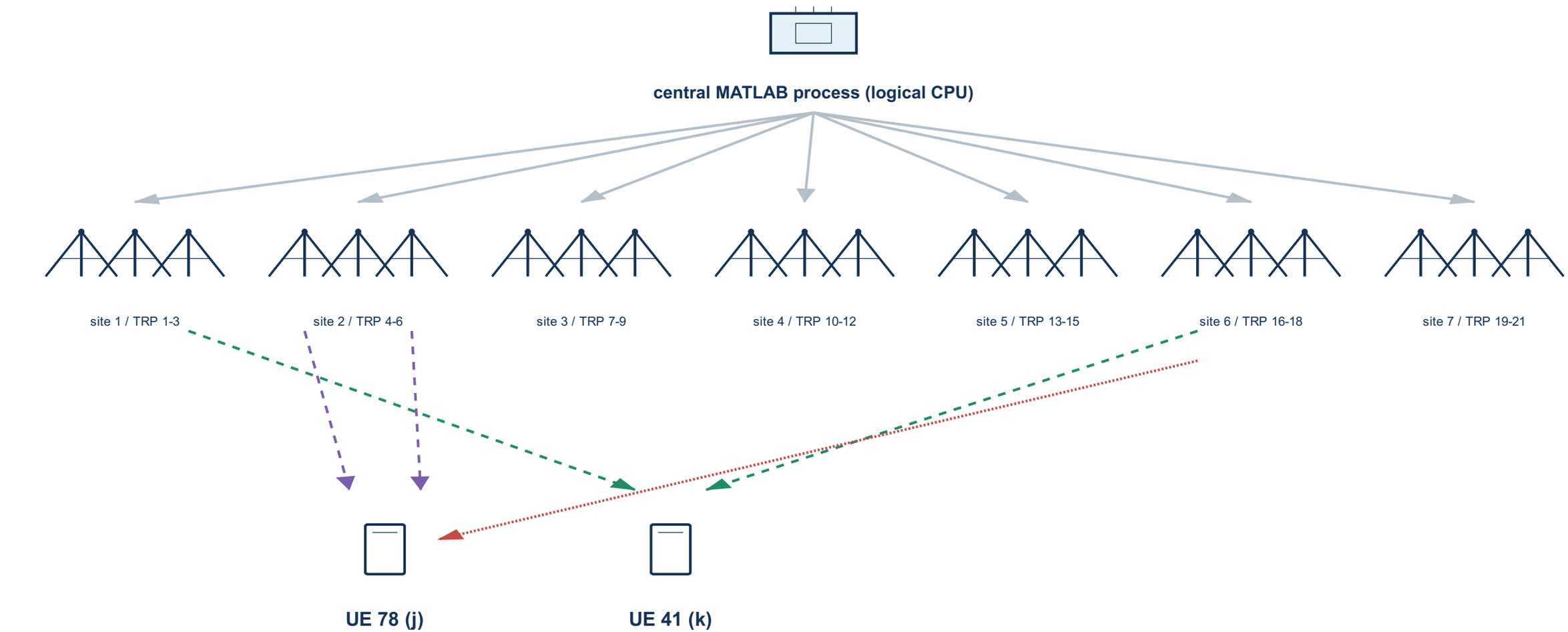
MATLAB source: matlab/chasi/ at 9527089; formula source: docs/CHASI_UNIFIED_TRACE.tex




This is execution order: scene.mat is written before solving; the optional dynamic replay runs after static evaluation and before final summary.

# 03 Seven-site / twenty-one-TRP topology

MATLAB source: matlab/chasi/ at 9527089; formula source: docs/CHASI_UNIFIED_TRACE.tex

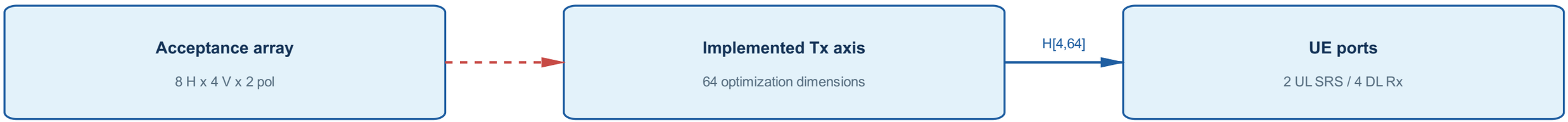



$$y_k = \sum_j \sum_l H^{\text{true}}_{klg} w_{ljg} s_j + n_k$$

Service IDs: saved 420-UE seed-20261011 group 3 manifest. Layout is schematic, NOT saved coordinates. Gray CPU-TRP lines are logical.

# 04 Antenna and matrix axes

MATLAB source: matlab/chasi/ at 9527089; formula source: docs/CHASI_UNIFIED_TRACE.tex




No array manifold or calibrated physical port mapping in the code

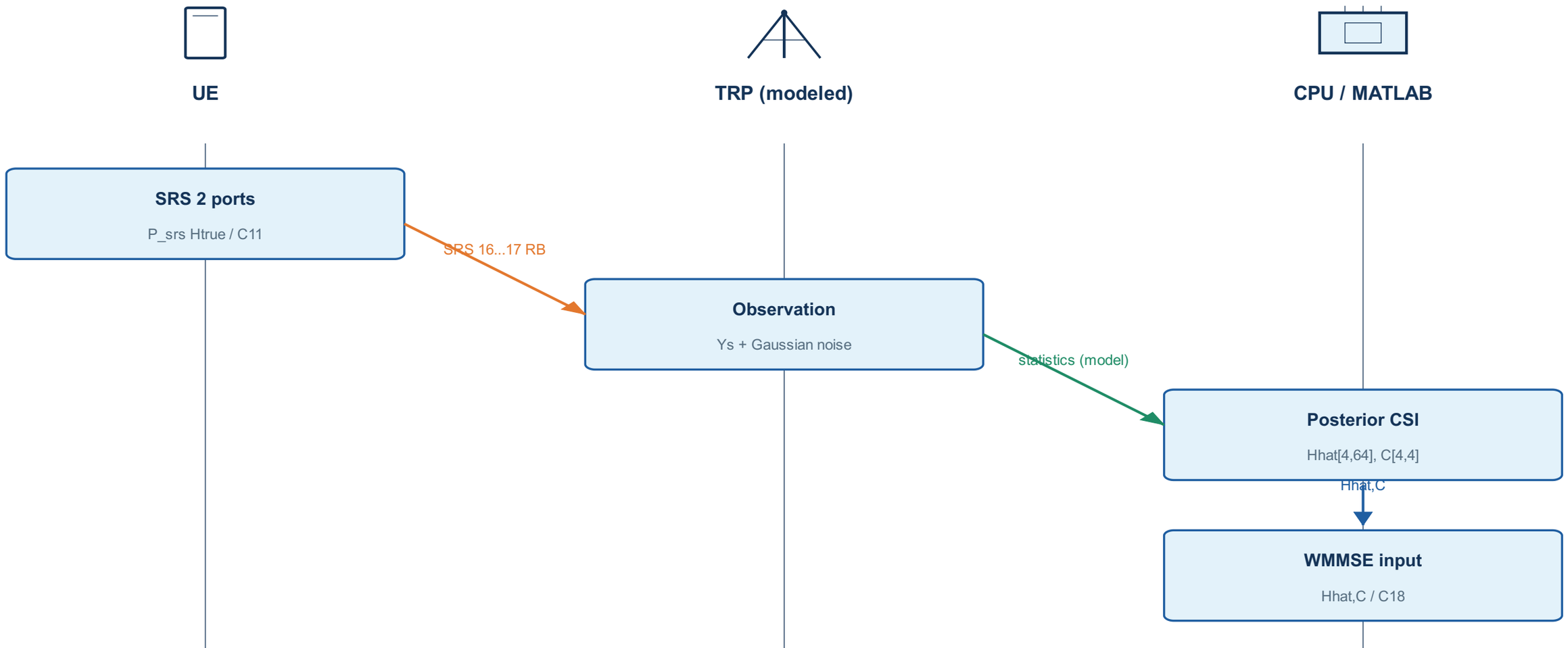
$$H_{klg} : 4 \times 64, \quad w_{ljg} : 64 \times r$$

$$P_{SRS} : 2 \times 4, \quad C_{klg} : 4 \times 4$$

C06/C11/C14: 4x64 correlated-Gaussian H does not implement 2D dual-polarized array response, reciprocity calibration, or port map.

# 05 UE / TRP / CPU: SRS and CSI

MATLAB source: matlab/chasi/ at 9527089; formula source: docs/CHASI_UNIFIED_TRACE.tex




17 hops x 20 TTI = 340 TTI; final hop 17 RB. No actual SRS waveform, IQ forwarding, or TRP-local estimator is simulated.

# 06 Scheduling, serving, rank, and power

MATLAB source: matlab/chasi/ at 9527089; formula source: docs/CHASI_UNIFIED_TRACE.tex



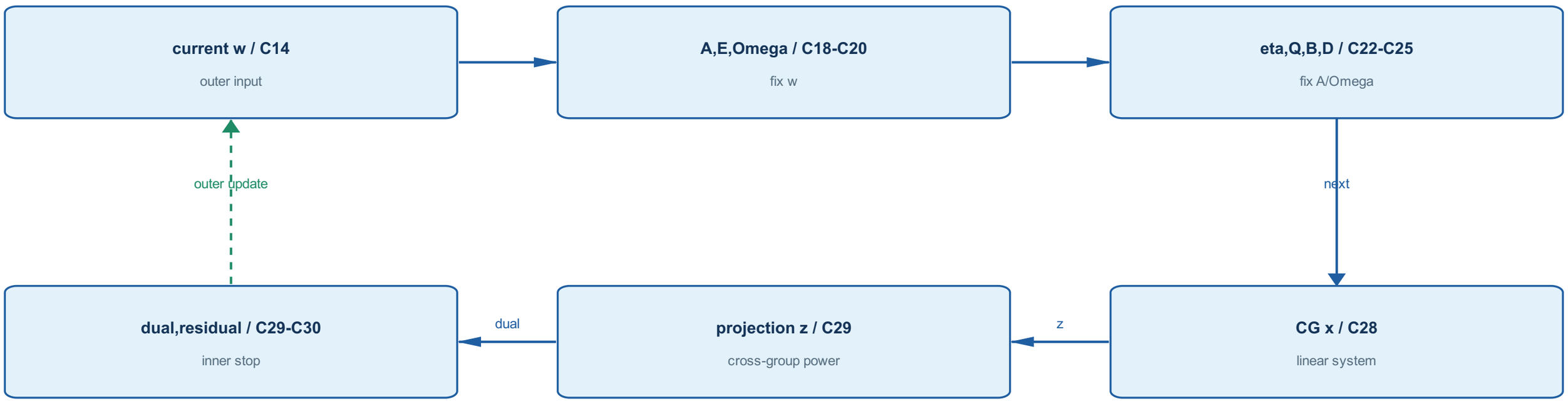

$$b_{kg} \in \{0,1\},\ \chi_{lkg} \in \{0,1\},\ \nu_g = n_g^{\mathrm{tx}}/273$$

---

nu_g = transmitted data RB / 273 enters objective, true rates and cross-group budget once; trafficLoad=.30 is separate.

# 07 WMMSE outer loop and centralized ADMM

MATLAB source: matlab/chasi/; formula source: docs/CHASI_UNIFIED_TRACE.tex

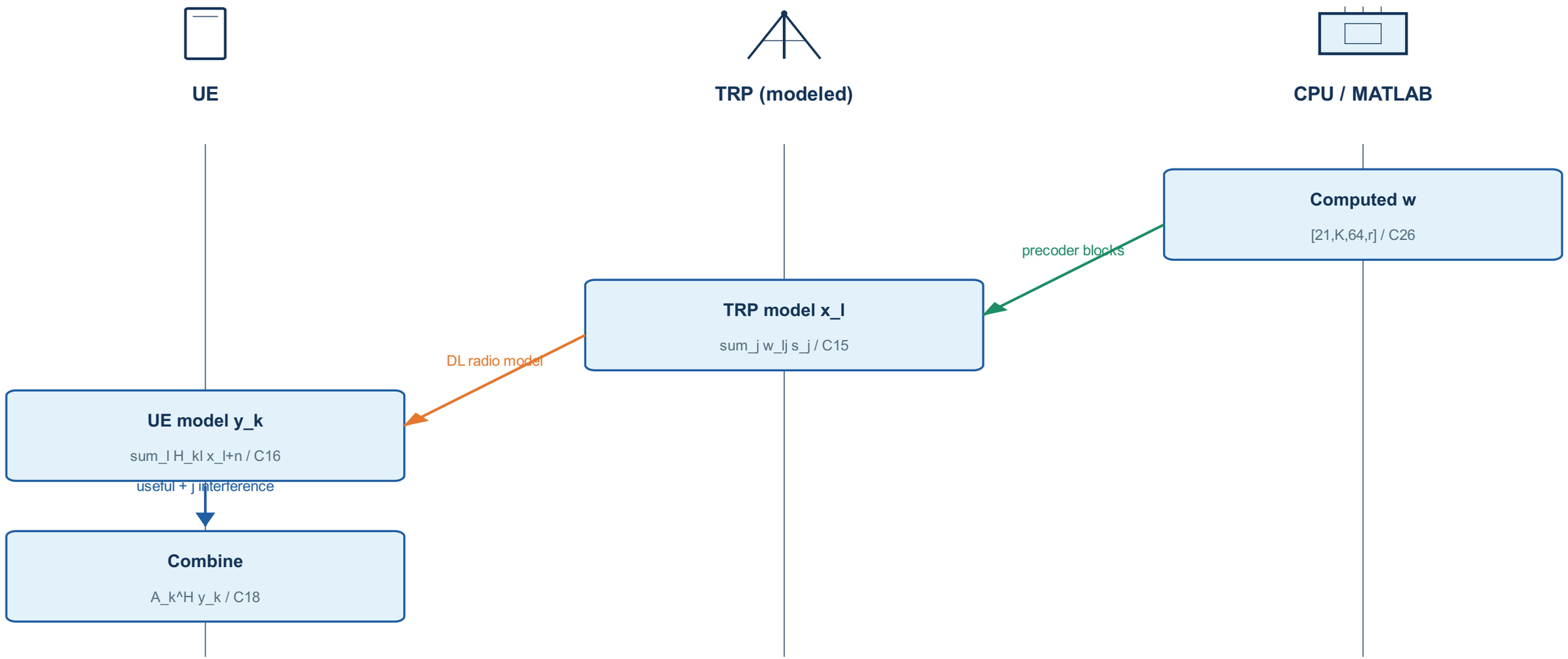



$$\mathrm{res}_{\mathrm{pri}} = \|x - z\|_F, \quad \mathrm{res}_{\mathrm{dual}} = \rho\|z - z^{\mathrm{old}}\|_F \quad \text{(C30)}$$

C30: primal and dual residuals must each meet epsilon plus CG test. Saved 210/420 traces reach inner cap 12 and outer cap 2, not tolerance stop; feasible z is returned.

# 08 CPU / TRP / UE: downlink model

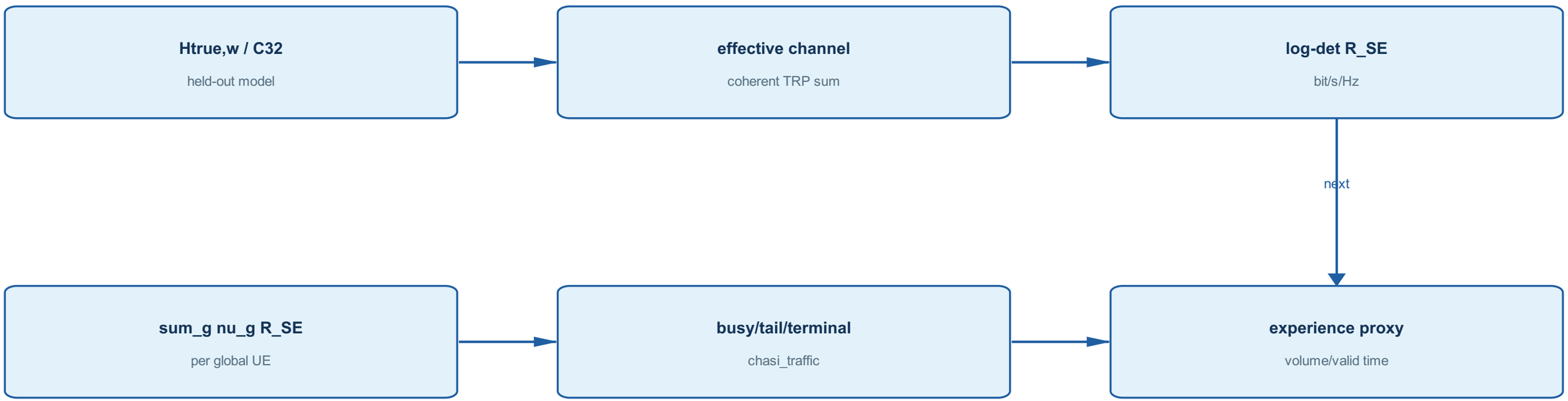
MATLAB source: matlab/chasi/ at 9527089; formula source: docs/CHASI_UNIFIED_TRACE.tex




Physical signal equations are model equivalents; the code computes complex covariances and log-det rates, not symbols or OFDM waveforms.

# 09 True-channel SE to experience proxy

MATLAB source: matlab/chasi/ at 9527089; formula source: docs/CHASI_UNIFIED_TRACE.tex



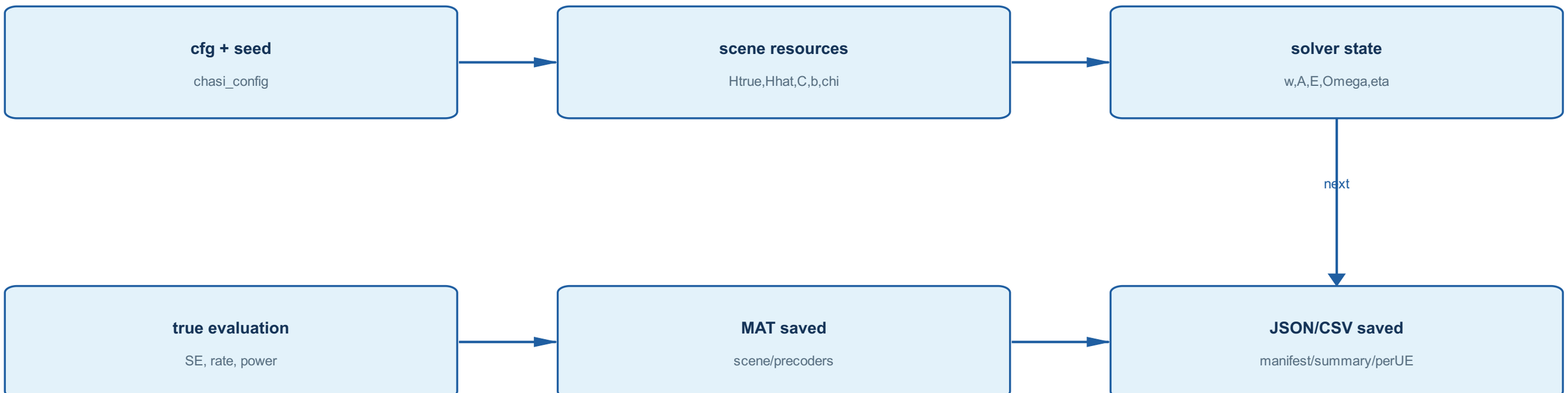

$$R_{\mathrm{SE},kg} = \log_2 \det(I + F_{kk} F_{kk}^H R_{I,k}^{-1})$$

---

No waveform or ACK/BLER. Valid-only and missing-as-zero bottom 5% are both reported; neither establishes formal TS28.554 acceptance.

# 10 Variable lifecycle and saved fields

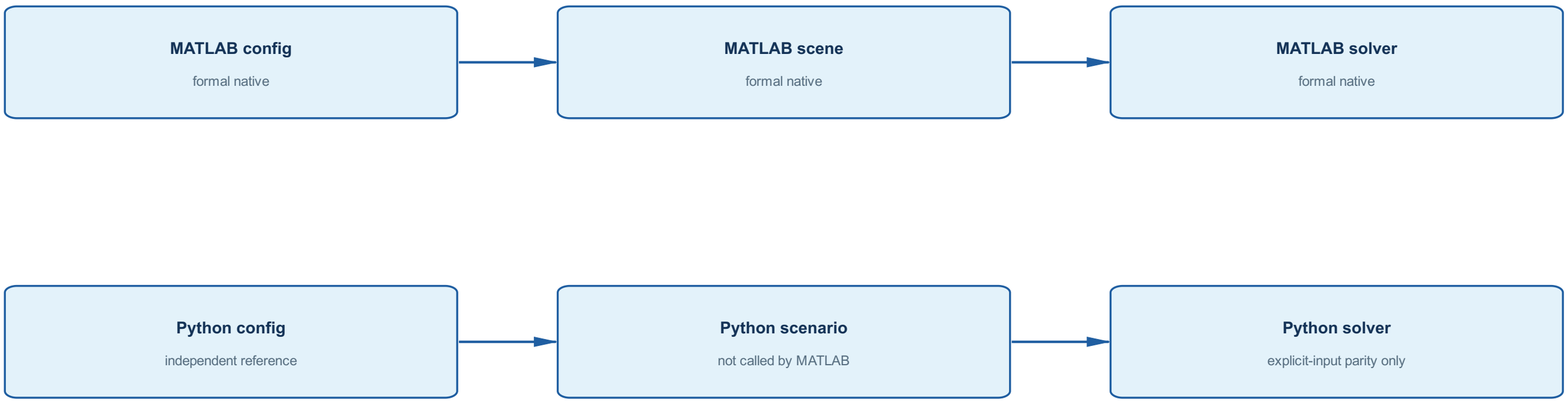
MATLAB source: matlab/chasi/ at 9527089; formula source: docs/CHASI_UNIFIED_TRACE.tex




CHASI v1/v2 fields convert only in chasi_load_precoder_result; legacy CF result.W is rejected. New schema separates precoder and mseWeight.

# 11 MATLAB formal chain and Python reference

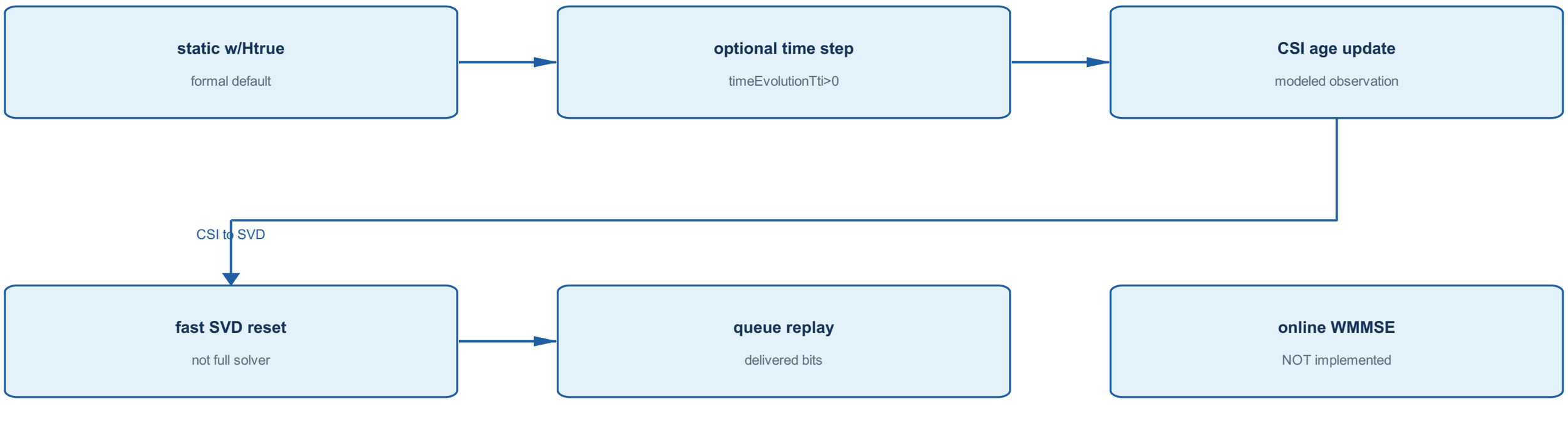
MATLAB source: matlab/chasi/; formula source: docs/CHASI_UNIFIED_TRACE.tex




No formal Python bridge. Five tiny fixtures pass; saved 420-UE full solve passes explicit-input parity. 210-UE final Omega misses absolute threshold by 1.98e-6.

# 12 Optional dynamic replay boundary

MATLAB source: matlab/chasi/; formula source: docs/CHASI_UNIFIED_TRACE.tex




$$H^{+} = \varphi\sqrt{\beta^{+}/\beta}H + \sqrt{(1-\varphi^{2})\beta^{+}}L_{\mathrm{rx}}Z$$

Replay step follows SRS hop interval (default 20 TTI). Saved 210-UE frozen-beam queue parity passes; mobility and online WMMSE remain unverified.